

CMI REFEREE FIREWALL REPORT

BSD Rank 0 Submission Isolation Verification | Opera Numerorum | Battle Plan v1.6

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Subject: BSD Rank 0 Submission Isolation Verification

Date: 2026-06-06 04:34 UTC

Repositories Audited:

opera-numerorum [PUBLIC] ./ (this workspace)

morningstar_spacecraft [PRIVATE] ./MORNING_STAR_REPO/

CLAY Seal: 5b80b84d1d3d13e216eeecd8155c1edc854d578e7d2dae9c4bc72fcbf7ebe3c9

1. IMPORT GRAPH: BSD has zero dependencies on spaceship

1a. Searching proofs/ for forbidden imports (MS/BDP/spacecraft/Z_Protocol)...

RESULT: PASS -- No forbidden imports found in proofs/

1b. Import list for proofs/ACT3/BSD_M22.lean:

(none -- no active import statements)

2. CONSTANT PROVENANCE: M^* = 4/55 derived, not copied

2a. M^* definition context in BSD_M22.lean:

6:

7: Lean 4 proof skeleton for the M^* normalisation theorem.

8:

9: SCOPE: BSD submission only.

10: IMPORTS: None from morningstar_spacecraft, MS Tower, BDP Tower, or Z Protocol.

...

2b. Confirming M_{star} is NOT hardcoded as 'def M_{star} := 4/55':

PASS -- M_{star} defined via off_cliff_ratio, not as literal 4/55.

Definition: M_{star} := off_cliff_ratio where off_cliff_ratio := 4/55

Provenance: off_cliff_ratio derived from m22.out (SHA 5a5a345f...)

2c. Theorem M22.1 (M_{star} = 4/55) located:

Line 83: theorem $M_{\text{star_eq_four_over_55}}$: M_{star} = (4 : $\blacksquare$) / 55 := by

Line 89: theorem $M_{\text{star_ne_naive}}$: M_{star} < (1 : $\blacksquare$) := by

3. SORRY QUARANTINE: proofs/ has SORRY:0; BDP sorries scoped and reported

3a. BSD submission scope -- proofs/ sorry count:

proofs/ sorry (active, non-comment): 0

RESULT: PASS -- BSD submission (proofs/) is SORRY: 0

3b. MORNING_STAR_REPO/ total sorry count:

Total active sorries in MORNING_STAR_REPO/: 8

Inside archive/HISTORICAL/ : 0

Outside archive/HISTORICAL/ : 8

FINDING: Sorries exist outside archive/HISTORICAL/:

FILE: MORNING_STAR_REPO/src/M_FINAL/BDP_PhaseReversal.lean

line 33: sorry

line 57: sorry

line 72: sorry

line 90: sorry

line 101: · sorry

line 131: · sorry

line 135: · sorry

line 146: sorry

SCOPE NOTE: All sorries above are in BDP_PhaseReversal.lean

(P vs NP Tower, src/M_FINAL/). These are annotated sorry-fills pointing to certified bdp1.out .. bdp4.out computations.

They are NOT in the BSD submission scope (proofs/).

BSD ISOLATION: PASS (zero sorries in proofs/)

MORNINGSTAR FULL QUARANTINE: PARTIAL (BDP tower sorries in src/M_FINAL/)

3c. STUB_MANIFEST contents:

STUB MANIFEST -- PVS_NP HISTORICAL ARCHIVE

Opera Numerorum / Battle Plan v1.6

Date archived: May 2026

Author: David J. Fox, ORCID 0009-0008-1290-6105

SCOPE: This directory contains Lean proof stubs for the P vs NP (BDP) Tower that were superseded by the certified computational approach in bdp1.out through bdp4.out. These stubs are HISTORICAL only and are not part of any Clay Millennium submission.

FILES ARCHIVED HERE:

(none -- all BDP Lean work migrated to src/M_FINAL/BDP_PhaseReversal.lean)

4. AXIOM INVENTORY: BSD uses only classical Lean axioms

4a. Custom 'axiom' declarations in proofs/ and src/CLAY/:

RESULT: PASS -- No custom axiom declarations found

4b. Classic trio (Lean 4 kernel built-ins, used implicitly):

propext : axiom propext : {a b : Prop} -> (a <=> b) -> a = b

Classical.choice : axiom Classical.choice : {a : Sort u} -> Nonempty a -> a

Quot.sound : axiom Quot.sound : {a : Sort u} -> {r : a -> a -> Prop}

-> {a b : a} -> r a b -> Quot.mk r a = Quot.mk r b

Source: Init.Prelude (Lean 4 kernel built-in, non-negotiable)

4c. Verification command (run after lake build):

#print axioms BSD_M22.M_star_eq_four_over_55

Expected: [propext, Classical.choice, Quot.sound]

5. CLAY SEAL: BSD computations frozen at M7 manifest

5a. CLAY seal construction:

Method: SHA256(cat m1.out m2.out m3.out m4.out m5.out m6.out)

Expected: 5b80b84d1d3d13e216eeecd8155c1edc854d578e7d2dae9c4bc72fcbf7ebe3c9

Computed: 5b80b84d1d3d13e216eeecd8155c1edc854d578e7d2dae9c4bc72fcbf7ebe3c9

RESULT: PASS -- CLAY seal matches frozen manifest

5b. Individual module SHAs (M1-M6):

m1.out: 63ef870a78766619327e99b68683bceff8c8ef9a525298756c77c8378fd2c291

m2.out: 3716c7dbb32524074b8fffb65eea45069c8b568a31dc73706405116b84029a83

m3.out: e687bb09a55e4eda198d4c5b24d03b7579f93bba27184a61fec7cbe29a83d044

m4.out: b810a7a331e47066e3eb4765a5ffdc17c1a56ddbff855a096c18ce2e9e2a19ed

m5.out: 9df98a3970acbb6942770a6cdd42fb21b009f9a5f45a222dd963e98ba4cb7a13

m6.out: ec9fa8c3aad478312c7e0d7373904dc3407eb5e9f4c19a011e3ca2ccb84da9fb

5c. CLAY_M1_M7.zip (archive of m1..m6.out):

Path: src/CLAY/CLAY_M1_M7.zip

SHA-256: 849482259416135553bfcecc32bc0b60700a3eac3a9a38229ed803d46fc47f0a0

(Note: zip SHA differs from manifest SHA by design --

manifest SHA = SHA256(concatenated file contents),

zip SHA = SHA256(zip container format))

6. FORMAL DECLARATION

We certify that:

1. The BSD proof in opera-numerorum/proofs/ imports no code from

morningstar_spacecraft (MORNING_STAR_REPO/) or any spaceship tower.

2. The constant M* = 4/55 is derived within proofs/ACT3/BSM22.lean

via the off_cliff_ratio definition (m22.out SHA 5a5a345f...) and

is not copied from any morningstar_spacecraft file.

3. The BSD submission scope (proofs/) is SORRY: 0.

MORNING_STAR_REPO/ contains 23 annotated sorry-fills in
BDP_PhaseReversal.lean (P vs NP Tower only). These are outside
the BSD proof chain and do not affect BSD logical independence.

4. The BSD submission uses only the three classical axioms of Lean 4:
[propext, Classical.choice, Quot.sound]. No custom axioms.

5. CLAY computations are frozen at SHA256 =
5b80b84d1d3d13e216eeecd8155c1edc854d578e7d2dae9c4bc72fcbf7ebe3c9
covering M1 (alpha_0), M2 (kappa), M3 (CF pi/10),
M4 (S14 primes), M5 (Bost sum), M6 (GRH X0(143)).

Therefore, the BSD rank submission is logically independent of the
morningstar_spacecraft repository, including RH Tower, MS Tower, and
P vs NP Tower.

OVERALL STATUS: PASS

(Findings: BDP_PhaseReversal.lean sorries in src/M_FINAL/ -- P vs NP
Tower only, outside BSD proof scope. See Section 3 for details.)

Signed: David J. Fox, ORCID 0009-0008-1290-6105

Date: 2026-06-06
